



FINDING MULTIPLE MIXED EQUILIBRIA

$$\begin{bmatrix} 2,1 & 3,6 & 5,3 \\ 3,6 & 1,1 & 1,2 \end{bmatrix} \begin{matrix} p \\ 1-p \end{matrix}$$

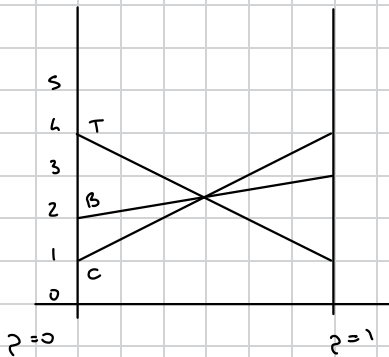
$q_1 \quad q_2 \quad q_3$

EXPECTED PAYOFF P2 : WE CHECK WHAT BOB CAN GAIN WITH ALICE'S MIXED STRATEGY

(L) $p + 4(1-p)$

(C) $4p + 1 \cdot (1-p)$

(R) $3p + 2(1-p)$



SINCE THEY ALL INTERSECT IN THE SAME POINT WE CAN CHOOSE ANY TWO LINES FOR THE INTERSECTION:

$$p + 4(1-p) = 4p + 1(1-p)$$

$$-3p + 4 = 3p + 1$$

$$6p = 3 \quad p = \frac{1}{2} \quad v = \frac{5}{2}$$

EXPECTED PAYOFF P1 : WE CHECK WHAT ALICE CAN GAIN WITH BOB'S MIXED STRATEGY

(T) $2q_1 + 3q_2 + 5q_3$

(B) $3q_1 + 1 \cdot q_2 + 1 \cdot q_3$

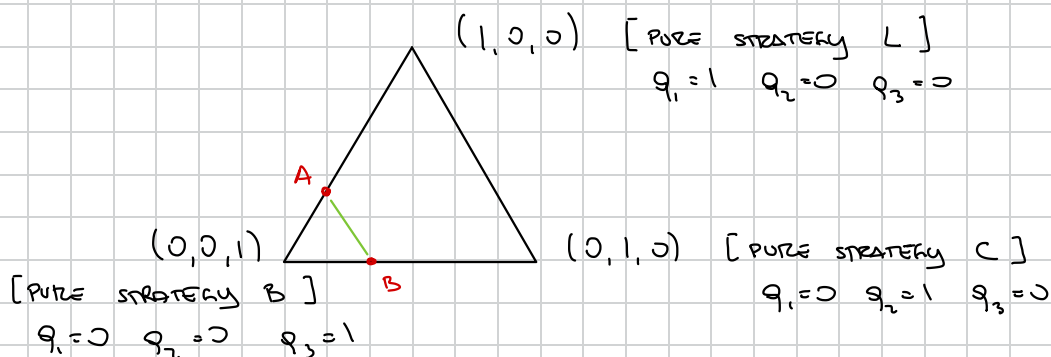
WE NEED TO FIND A MIXED STRATEGY WHICH IS INDIFFERENT TO WHAT ALICE DO :

$$2q_1 + 3q_2 + 5q_3 = 3q_1 + q_2 + q_3$$

$$-q_1 + 2q_2 + 4q_3 = 0$$

(*) $\begin{cases} q_1 = 2q_2 + 4q_3 \\ q_1 = 1 - q_2 - q_3 \end{cases}$ INFINITE POSSIBILITIES! ALL THE ONES THAT SATISFY THIS SYSTEM OF EQUATIONS!

IF WE WANT TO BE MORE SPECIFIC, THIS IS A SEGMENT [CONVEX COMBINATION] BETWEEN TWO MIXED PROBABILITIES!



WE CAN FIND THIS POINTS ON THE EDGES OF THE TRIANGLE [i.e. BY SETTING ONE OF THE PROBABILITIES TO ZERO] AND ENSURING WE THEY SATISFY THE EQUATIONS (*)

$$\begin{array}{ll} q_1 = 0 & 2q_2 = -4q_3 \quad \text{IMPOSSIBLE!} \\ q_2 = 0 & \begin{cases} q_1 = 4q_3 \\ q_1 = 1 - q_3 \end{cases} \quad q_1 = \frac{4}{5} \quad q_3 = \frac{1}{5} \quad \text{A} \\ q_3 = 0 & \begin{cases} q_1 = 2q_2 \\ q_1 = 1 - q_2 \end{cases} \quad q_1 = \frac{2}{3} \quad q_2 = \frac{1}{3} \quad \text{B} \end{array}$$